CLOUD AND FOG COMPUTING · REPORT

Amazon Web Services (AWS)

What it is, how it works and why it matters



Polina Chernikova · Group 1307 · November 2026

CLOUD BASICS

Own servers vs the cloud

Topic	Own servers (on-premises)	Cloud (AWS)
Start cost	Big payment for hardware	Almost zero to start
Time to start	Weeks or months	A few minutes
Need more power?	Buy and install new machines	Add more with a few clicks
Repairs and updates	Your own IT team	AWS takes care of hardware
Payment	Fixed, even when unused	Only for what you use

MEET AWS

Amazon Web Services in short

- Started by Amazon in **2006**
- First services: **S3** (file storage) and **EC2** (virtual servers)
- Today: **more than 200 services** — from servers to AI
- Millions of customers: startups, banks, governments, universities

2006
year AWS started

200+
services

39
Regions

123
Availability Zones

Source: aws.amazon.com, November 2026

MARKET

AWS is the market leader

Share of the cloud infrastructure market, July–September 2026



≈ $46 billion

AWS revenue in one quarter — 39% more than a year before

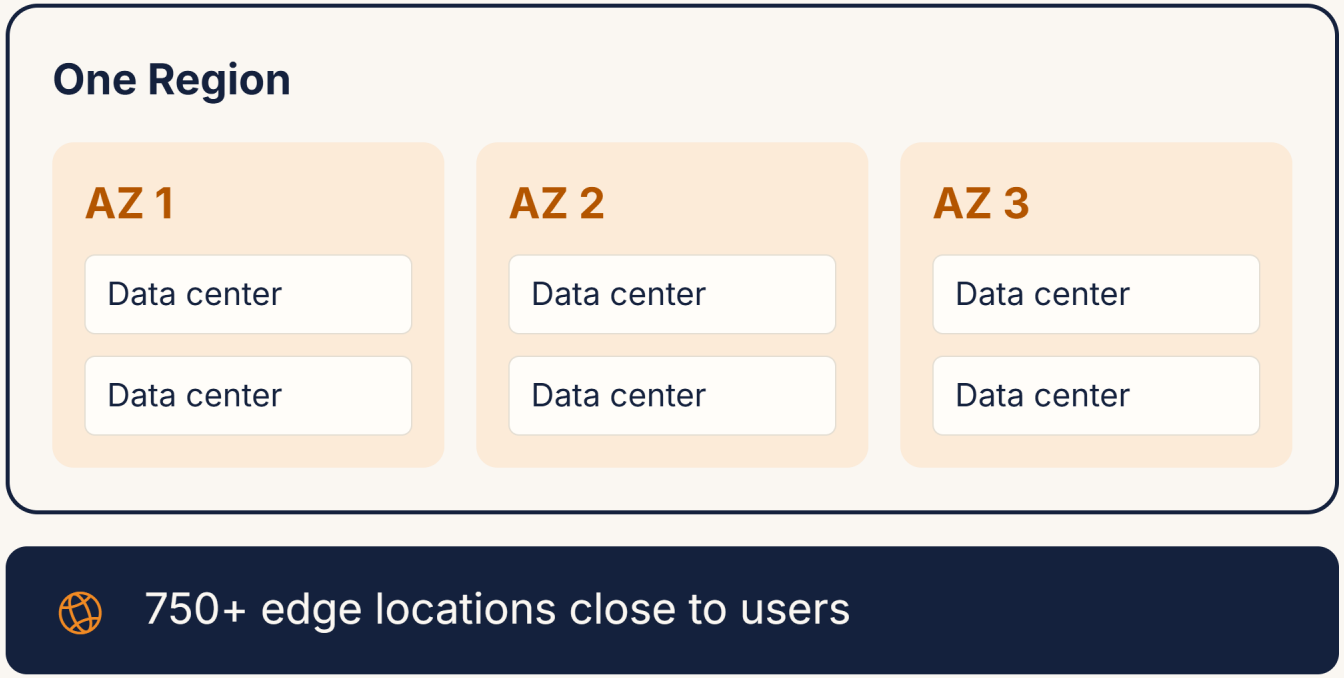
+48% and +103%

Growth of Azure and Google Cloud — faster than AWS

Sources: Synergy Research Group; company results for Q3 2026

GLOBAL INFRASTRUCTURE

How AWS is built around the world



Region

A big geographic area with several AZs. 39 Regions today.

Availability Zone (AZ)

One or more data centers with their own power and network. 123 in total.

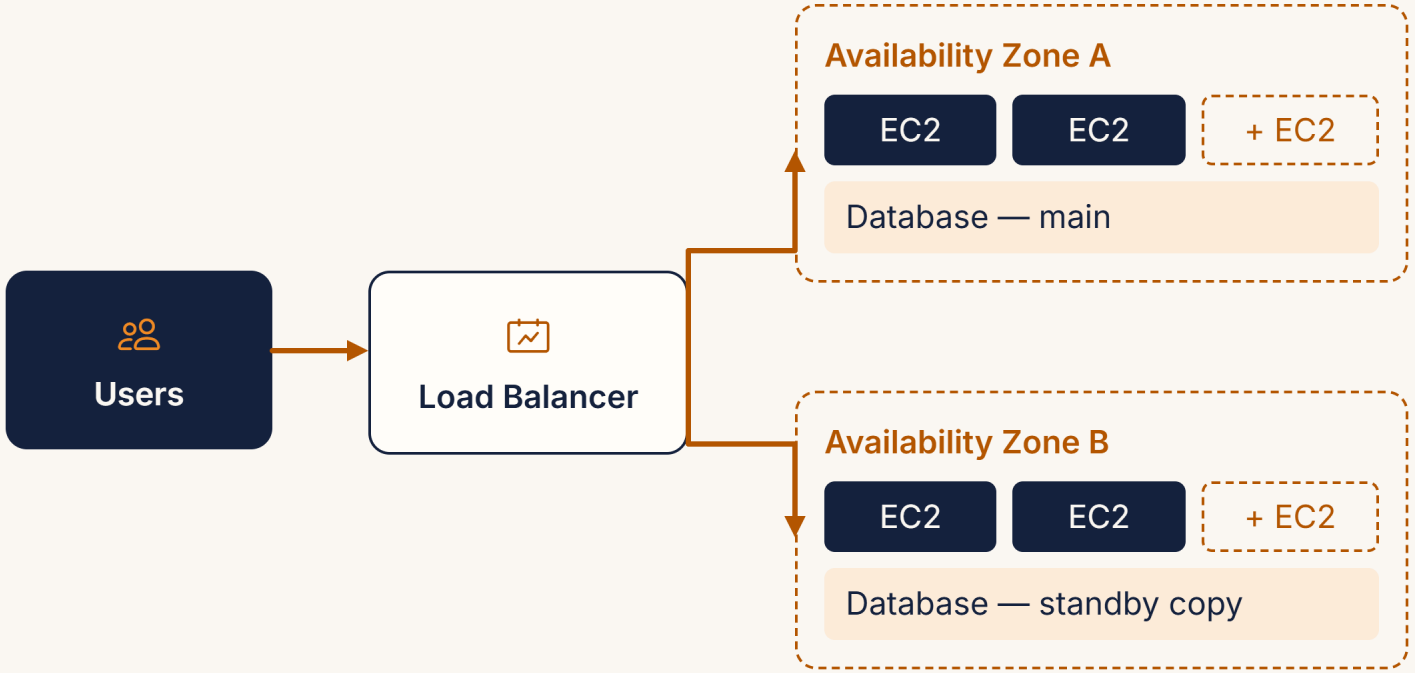
Edge location

A small site in a city that keeps copies of popular content.

Source: aws.amazon.com/about-aws/global-infrastructure, November 2026

RELIABILITY AND SCALE

How AWS keeps an app running



Elastic Load Balancing

Spreads traffic across servers in several AZs.

Auto Scaling

Adds servers when traffic grows, removes them when it falls.

Multi-AZ database

A standby copy takes over if the main one fails.

IAAS · PAAS · SAAS

Three service models

IaaS

Infrastructure as a Service

You rent servers, storage and network. You set up the rest.

You manage: OS, apps, updates

AWS example: Amazon EC2

PaaS

Platform as a Service

You bring only your code. The platform runs it for you.

You manage: only your code

AWS example: Elastic Beanstalk

SaaS

Software as a Service

A ready app you simply use, often in the browser.

You manage: only your data and users

Example: Gmail, Microsoft 365

You manage more → Provider manages more

MAIN SERVICES

Six services to know first

Amazon EC2

COMPUTE

Virtual servers. Start one in minutes.

AWS Lambda

SERVERLESS

Runs code with no server to manage.

Amazon S3

STORAGE

Stores any files: photos, videos, backups.

Amazon RDS

SQL DATABASE

Classic databases, managed by AWS.

Amazon DynamoDB

NOSQL DATABASE

Very fast database for big apps.

Amazon CloudFront

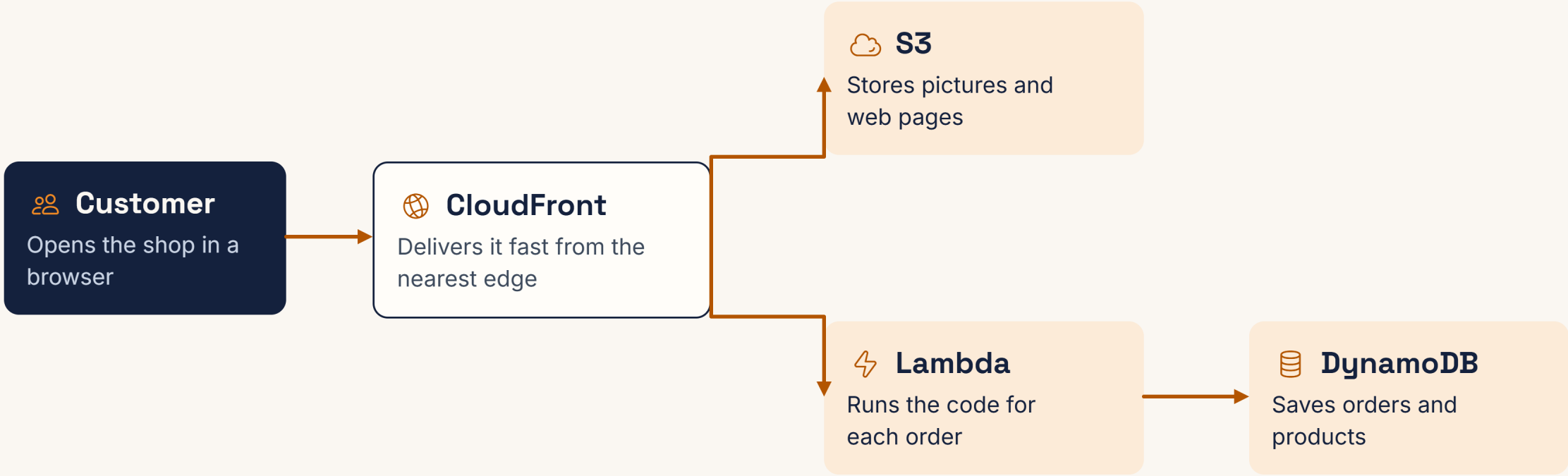
DELIVERY (CDN)

Delivers content fast, close to users.

PUTTING IT TOGETHER

How the services work together

Example: a small online shop on AWS — with no server to manage



SECURITY

Security is a shared job

AWS protects the cloud

- Buildings and guards
- Hardware and network
- Basic software that runs it all

You protect what you put in it

- Your data and backups
- Users and passwords (IAM)
- Settings and network rules (VPC)

IAM — Identity and Access Management: decides who can do what.

VPC — Virtual Private Cloud: your own private network inside AWS.

ARTIFICIAL INTELLIGENCE

AI on AWS

1/3 of the whole cloud market is now AI services

Amazon Bedrock

Use ready AI models — chat, text, images — through an API. No training needed.

Amazon SageMaker

Build and train your own machine learning models.

Source: Synergy Research Group, Q3 2026

CLOUD AND FOG

Cloud vs fog computing

Cloud
AWS Regions: huge power, but far away

only short reports go up

Fog
Gateways and small servers: close and fast

raw data from devices

Devices
Sensors, cameras, cars, machines

Fog = computing between devices and the cloud, close to the data.

AWS TOOLS FOR THE EDGE

AWS IoT Greengrass
Runs AWS code on local devices

AWS Outposts
AWS hardware in your own building

Local Zones (43)
AWS close to big cities

Wavelength Zones (33)
AWS inside 5G mobile networks

Source: aws.amazon.com/about-aws/global-infrastructure, November 2026

PRICES

How you pay for AWS

Pay-as-you-go: no upfront cost, billed by the second, hour or gigabyte.

Model	How it works
On-Demand	Pay per second or hour. No promise.
Savings Plans	Promise 1–3 years of use, pay much less.
Spot	Spare servers, up to 90% cheaper. AWS can take them back.

Free Tier for new accounts

- $100 credits at sign-up
- Up to $100 more as you try services
- Free plan: up to 6 months

Tip: set a budget alert in AWS Budgets.

Source: aws.amazon.com/free, November 2026

REAL LIFE

Who uses AWS and what for

FAMOUS EXAMPLE

Netflix

On AWS since about 2009. AWS servers, storage and data tools help it serve 300+ million subscribers.

- Websites and apps
- Video and games
- Data analysis
- AI and chatbots
- Backup and storage
- Science and research

Also on AWS: Airbnb, many banks, governments, universities and thousands of startups.

PROS AND CONS

Strengths and weaknesses

Strengths

- Start fast: no hardware to buy
- Scale up and down easily
- Pay only for what you use
- Data centers all over the world
- 200+ services in one place

Weaknesses

- Complex: many services to learn
- Bills can surprise you
- Vendor lock-in: hard to move away
- Outages happen (October 2025)

DEMO · ABOUT 5 MINUTES

How AWS Lambda works

A small web page made by code — with no server

1. Create a function in the AWS console
2. Paste the Python code
3. Click Deploy
4. Turn on a Function URL

DEMO · THE CODE

17 lines of Python

```
from datetime import datetime, timezone
from html import escape

def lambda_handler(event, context):
    # 1. Read the name from the web address: ...?name=Polina
    params = event.get("queryStringParameters") or {}
    name = escape(params.get("name", "students"))
    # 2. Find out the time and the ID of this call
    time = datetime.now(timezone.utc).strftime("%H:%M:%S UTC")
    request_id = context.aws_request_id[:8]
    # 3. Build a small web page (HTML)
    page = (f"<h1>Hello, {name}!</h1>"
            f"<p>Made by AWS Lambda at {time}, request {request_id}</p>")
    # 4. Send the page back to the browser
    return {"statusCode": 200,
            "headers": {"Content-Type": "text/html; charset=utf-8"},
            "body": page}
```

1. Read the name from the address
2. Get the time and the call ID
3. Build the web page
4. Send it to the browser

No server code — AWS takes care of it.

DEMO · HOW IT RUNS

From a link to a web page

1. A person opens the Function URL
2. AWS starts the code in milliseconds
3. The code returns the page
4. The browser shows it



EXAMPLE RESULT

Refresh the page — the time and the request ID change.

Nobody opens it — the code does not run, and you pay nothing.

SUMMARY

Key takeaways

1. Load balancing, Auto Scaling and Multi-AZ keep apps running.
2. AWS is the biggest cloud provider: 200+ services, 39 Regions.
3. Fog and edge computing bring the cloud closer to devices.
4. Start small: the Free Tier lets you try AWS at no cost.